

cs224n a2 mzx

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(a)

$$- \sum_{w \in \text{Vocab}} y_w \log(\hat{y}_w) = -(y_o \log(\hat{y}_o) + \sum_{w \in \text{Vocab}, w \neq o} y_w \log(\hat{y}_w)) = -\log(\hat{y}_o),$$

where $y_o = 1$, $y_w = 0$ for $w \neq o$.

(b)

$$\begin{aligned}\frac{\partial J_{\text{naive-softmax}}(v_c, o, U)}{\partial v_c} &= -\frac{\partial}{\partial v_c} \log \mathbb{P}(O = o \mid C = c) \\&= -\frac{\partial}{\partial v_c} \log \frac{\exp(u_o^\top v_c)}{\sum_{w \in \text{Vocab}} \exp(u_w^\top v_c)} \\&= -\frac{\partial}{\partial v_c} \left[\log \exp(u_o^\top v_c) - \log \sum_{w \in \text{Vocab}} \exp(u_w^\top v_c) \right] \\&= -\frac{\partial}{\partial v_c} \left[u_o^\top v_c - \log \sum_{w \in \text{Vocab}} \exp(u_w^\top v_c) \right] \\&= -u_o + \frac{\partial}{\partial v_c} \left[\log \sum_{w \in \text{Vocab}} \exp(u_w^\top v_c) \right] \\&= -u_o + \frac{1}{\sum_{w \in \text{Vocab}} \exp(u_w^\top v_c)} \frac{\partial}{\partial v_c} \left[\sum_{w \in \text{Vocab}} \exp(u_w^\top v_c) \right] \\&= -u_o + \frac{1}{\sum_{w \in \text{Vocab}} \exp(u_w^\top v_c)} \left[\sum_{w \in \text{Vocab}} \exp(u_w^\top v_c) \frac{\partial}{\partial v_c} (u_w^\top v_c) \right] \\&= -u_o + \frac{1}{\sum_{w \in \text{Vocab}} \exp(u_w^\top v_c)} \left[\sum_{w \in \text{Vocab}} \exp(u_w^\top v_c) u_w \right] \\&= -u_o + \sum_{w \in \text{Vocab}} \frac{\exp(u_w^\top v_c) u_w}{\sum_{x \in \text{Vocab}} \exp(u_x^\top v_c)} \\&= -u_o + \sum_{w \in \text{Vocab}} \mathbb{P}(O = w \mid C = c) u_w \\&= -u_o + \sum_{w \in \text{Vocab}} \hat{y}_w u_w \\&= -y_o u_o + \sum_{w \in \text{Vocab}} \hat{y}_w u_w \\&= -\sum_{x \in \text{Vocab}} y_x u_x + \sum_{w \in \text{Vocab}} \hat{y}_w u_w \\&= \sum_{x \in \text{Vocab}} (\hat{y}_x - y_x) u_x \\&= U(\hat{y} - y)\end{aligned}$$

(c)

Case 1: $w \neq o$

$$\begin{aligned}
\frac{\partial J_{\text{naive-softmax}}(v_c, o, U)}{\partial u_w} &= -\frac{\partial}{\partial u_w} \left[u_o^\top v_c - \log \sum_{x \in \text{Vocab}} \exp(u_x^\top v_c) \right] \\
&= \frac{\partial}{\partial u_w} \left[\log \sum_{x \in \text{Vocab}} \exp(u_x^\top v_c) \right] \\
&= \frac{1}{\sum_{x \in \text{Vocab}} \exp(u_x^\top v_c)} \frac{\partial}{\partial u_w} \left[\sum_{x \in \text{Vocab}} \exp(u_x^\top v_c) \right] \\
&= \frac{1}{\sum_{x \in \text{Vocab}} \exp(u_x^\top v_c)} \left[\sum_{x \in \text{Vocab}} \frac{\partial}{\partial u_w} \exp(u_x^\top v_c) \right] \\
&= \frac{1}{\sum_{x \in \text{Vocab}} \exp(u_x^\top v_c)} \left[\sum_{x \in \text{Vocab}} \exp(u_x^\top v_c) \frac{\partial}{\partial u_w} (u_x^\top v_c) \right] \\
&= \frac{1}{\sum_{x \in \text{Vocab}} \exp(u_x^\top v_c)} \exp(u_w^\top v_c) \frac{\partial}{\partial u_w} (u_w^\top v_c) \\
&= \frac{\exp(u_w^\top v_c)}{\sum_{x \in \text{Vocab}} \exp(u_x^\top v_c)} v_c \\
&= \hat{y}_w v_c
\end{aligned}$$

Case 2: $w = o$

$$\begin{aligned}
\frac{\partial J_{\text{naive-softmax}}(v_c, o, U)}{\partial u_w} &= -\frac{\partial}{\partial u_w} \left[u_o^\top v_c - \log \sum_{x \in \text{Vocab}} \exp(u_x^\top v_c) \right] \\
&= -\frac{\partial}{\partial u_w} (u_o^\top v_c) + \hat{y}_w v_c \\
&= -v_c + \hat{y}_w v_c \\
&= (\hat{y}_w - 1) v_c
\end{aligned}$$

Thus,

$$\frac{\partial J_{\text{naive-softmax}}}{\partial u_w} = \begin{cases} (\hat{y}_o - 1) v_c, & w = o \\ \hat{y}_w v_c, & w \neq o \end{cases}$$

Equivalently, since $y_o = 1$ and $y_w = 0$ ($w \neq o$), this can be written as:

$$\frac{\partial J_{\text{naive-softmax}}}{\partial u_w} = (\hat{y}_w - y_w) v_c.$$

(d)

$$\sigma(x) = \frac{1}{1 + \exp(-x)}$$